Sanghoon Lee

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Research Group for Bio-Embodied Physical AI
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🌐 https://huno130.github.io/
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> *I unify AI, middleware, and control to realize Embodied Intelligence under uncertainty and resource constraints.*

For full publications, open-source projects, talks, and an interactive research map, see huno130.github.io.

RESEARCH INTERESTS

Cyber-Physical Artificial Intelligence (CPAI) — unifying AI, middleware, and control to realize Embodied Intelligence under uncertainty and resource constraints. Particular depth in the **ROS 2 core and its middleware internals** (DDS — FastDDS, CycloneDDS, RTI Connext — and Zenoh): QoS semantics, discovery (SPDP/SEDP), the RTPS wire protocol, executors, and observability.

- **AI**. Physical AI, Multi-Robot Systems, Autonomous Mobile Robots, Deep Reinforcement Learning
- **ROS 2**. Core / middleware internals (RTPS, DDS, Zenoh), QoS & Discovery, Observability, Next-gen Middleware, Cross-layer Optimization, Software-Defined Robotics
- **CPS**. Wireless Control Systems, Cyber-Physical Security, Resilient Architectures, Vulnerability Analysis

EDUCATION

Period	Institution	Degree / Program
2022.02 – 2026.02	DGIST, Daegu, South Korea	Ph.D. in Electrical Engineering and Computer Science Combined Program (UG-G.). Major GPA 4.3 / 4.3 Advisor. Prof. Kyung-Joon Park
2019.06 – 2019.09	University of California, Los Angeles (UCLA), USA	International Student Program. Major GPA 4.3 / 4.3
2018.03 – 2022.02	DGIST, Daegu, South Korea	B.S. in Computer Science. Major GPA 3.99 / 4.3

EXPERIENCE

Period	Role	Affiliation
2026.02 – Present	InnoCore Postdoctoral Research Fellow	Research Group for Bio-Embodied Physical AI, DGIST
2022.02 – 2026.02	Graduate Research Assistant	Cyber-Physical Systems Integration (CSI) Lab, DGIST
2020.08 – 2022.02	Undergraduate Research Student	Cyber-Physical Systems Integration (CSI) Lab, DGIST

Period	Role	Affiliation
2020.01 – 2020.02	Research Intern (Winter)	Computer Architecture and Systems (CAS) Lab, DGIST

PH.D. DISSERTATION

Cyber-Physical Artificial Intelligence. A Unified Framework for Embodied Intelligence
Sanghoon Lee. Advisor. Prof. Kyung-Joon Park.
DGIST, Feb. 2026.
https://scholar.dgist.ac.kr/handle/20.500.11750/59636

INTERNATIONAL PUBLICATIONS

> *33+ peer-reviewed papers, including 19 first or co-first authored.*
> ** indicates equal contribution as first authors.*

Under Review · In Preparation (2026)

[U13]**Cyber-Physical AI. Reconstructing Cyber-Physical Systems in the Age of Physical Artificial Intelligence**
Sanghoon Lee, Guoliang Xing, Xue Liu, Karl H. Johansson, Kyung-Joon Park.
Invited Paper. ACM Transactions on Cyber-Physical Systems (TCPS), 20th Anniversary Special Issue (TCPS-2026-0101). *Under Review*.

[U12]**Harness Engineering for Physical AI. Robot Middleware Is the Harness Layer**
Sanghoon Lee, Jiyeong Chae, Kyung-Joon Park.
ACM/IFIP/USENIX International Middleware Conference (Middleware) 2026. Big Ideas Track (#256). *Under Review*. arXiv:2606.09416.

[U11]**ros2probe. Non-intrusive, Kernel-selective Observability for Robot Operating System 2 Middleware**
Jisang Yu*, **Sanghoon Lee***, Yeonwoo Choi, Kyung-Joon Park.
ACM/IFIP/USENIX International Middleware Conference (Middleware) 2026. Research Paper (#228). *Under Review*. arXiv:2606.10746 · Project Page.

[U10]**The Three Dimensions of ROS 2 Middleware**
Sanghoon Lee, Taehun Kim, Angelo Corsaro, Kyung-Joon Park.
IEEE Communications Surveys & Tutorials (COMST). *Under Review*.

[U9]**From Hype to Reality. A Survey of Physical AI in Industrial Robotics**
Sanghoon Lee*, Jiyeong Chae*, Jinhong Park, Kyung-Joon Park.
Engineering Applications of Artificial Intelligence (EAAI). *Under Review*.

[U8]**Dependency Chain Analysis of ROS 2 DDS QoS Policies. From Lifecycle Tutorial to Static Verification**
Sanghoon Lee, Junha Kang, Kyung-Joon Park.
IEEE Internet of Things Journal (IoT-J). *2nd Minor Revision* (IoT-62530-2026.R1). arXiv:2509.03381.

[U7]**Probabilistic Latency Analysis of the Data Distribution Service in ROS 2**
Sanghoon Lee, Hyung-Seok Park, Jiyeong Chae, Kyung-Joon Park.
IEEE Transactions on Vehicular Technology (TVT). *Under Review*. arXiv:2508.10413.

[U6]**Optimizing ROS 2 Communication for Wireless Robotic Systems**
Sanghoon Lee, Taehun Kim, Yeonwoo Choi, Kyung-Joon Park.
IEEE Internet of Things Journal (IoT-J). *Under Review*. arXiv:2508.11366.

[U5]**LIMA. Local Intersection Marshaling Algorithm for Multi-Agent Path Finding in Large-Scale Industrial Environments**

Sanghoon Lee*, Jisang Yu*, Geon-pyo Kim, Kyung-Joon Park.
Robotics and Autonomous Systems (RAS), Elsevier. *Under Review*. SSRN.

[U4] **Exploiting ROS 2's Blind Spot. Cross-Layer Attacks from Discovery to ARP Spoofing**
Hyunho Ryu*, **Sanghoon Lee***, Young-Sik Kim, Dan Dongseong Kim, Kyung-Joon Park.
IEEE Internet of Things Journal (IoT-J). *Under Review* (IoT-67776-2026).

[U3] **Natural-Language to Operational Policies. LLM-Driven Costmaps for Industrial AMRs**
Jiyeong Chae, Hyunkyo Seo, Heonjae Lee, **Sanghoon Lee**, Kyung-Joon Park.
Robotics and Autonomous Systems (RAS). *Under Review*.

[U2] **Robust Localization in Dynamic Indoor Environments Using Structure-Guided Point-wise LiDAR Scoring**
Yeong-gi Hong*, Donghyung Lee, **Sanghoon Lee**, Jiyeong Chae, Kyung-Joon Park.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026. *Under Review*.

[U1] **Mitigating DDS Discovery Bursts in ROS 2 Multi-Robot Systems**
Taehyun Kim, **Sanghoon Lee**, Kyung-Joon Park.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026. *Under Review*.

[IP5] **Analytical Modeling of Discovery Storms in DDS and Zenoh for Large-Scale Robotic Networks**
Yeonwoo Choi*, **Sanghoon Lee***, Kyung-Joon Park.
Target. IEEE INFOCOM 2027. *In Preparation* (submission 2026-07).

Published

[19] **Discovery Storm. Scalability Analysis of DDS and Zenoh in Large-Scale Wireless Robotic Networks**
Sanghoon Lee, Yeonwoo Choi, Jiyeong Chae, Kyung-Joon Park.
IEEE INFOCOM NetRobiCS 2026 Workshop, Feb. 2026.

[18] **Time-aware Costmap for Smoother and Less Disruptive AMR Navigation With ROS 2**
Jiyeong Chae, Hyunkyo Seo, **Sanghoon Lee**, Kyung-Joon Park.
International Journal of Control, Automation, and Systems (IJCAS), Dec. 2025.
https://link.springer.com/article/10.1007/s12555-025-0558-8

[17] **DIDoS. Disturbance-induced Denial-of-Service Attack in Networked Cyber-Physical Systems**
Sangjun Kim, **Sanghoon Lee**, Kyung-Joon Park.
International Journal of Control, Automation, and Systems (IJCAS), Nov. 2025.
https://link.springer.com/article/10.1007/s12555-025-0572-x

[16] **Vulnerability Analysis of Cross-Layer Attacks on ROS 2**
Hyunho Ryu, **Sanghoon Lee**, Kyung-Joon Park.
International Conference on ICT Convergence (ICTC) 2025, Oct. 2025.
https://csi.dgist.ac.kr/HOME/csi/data/upload/20251016214727680e95fa4555.pdf

[15] **Poster. How to Send Large Data in ROS 2**
Sanghoon Lee, Taehun Kim, Jiyeong Chae, Kyung-Joon Park.
IEEE International Conference on Network Protocols (ICNP) 2025, Sep. 2025.
https://ieeexplore.ieee.org/abstract/document/11192433/

[14] **From Issues to Routes. A Cooperative Costmap with Lifelong Learning for Multi-AMR Navigation**
Jiyeong Chae*, **Sanghoon Lee***, Hyunkyo Seo, Kyung-Joon Park.
Journal of Industrial Information Integration (JIII), Sep. 2025.
https://www.sciencedirect.com/science/article/abs/pii/S2452414X25001645

[13] **Integrating ROS 2 and Physical AI. Architecture and Challenges**
Sanghoon Lee, Jiyeong Chae, Kyung-Joon Park.

International Conference on Ubiquitous and Future Networks (ICUFN) 2025, Jul. 2025.
https://ieeexplore.ieee.org/abstract/document/11170038/

[12] **An Analytical Latency Model of the Data Distribution Service in ROS 2**
Hyung-Seok Park, **Sanghoon Lee**, Kyung-Joon Park.
IEEE International Conference on Computer Communications (INFOCOM) 2025, May 2025.
https://ieeexplore.ieee.org/abstract/document/11044454/

[11] **PINMAP. A Cost-efficient Algorithm for Glass Detection and Mapping Using Low-cost 2D LiDAR**
Jiyeong Chae, Hyunkyo Seo, **Sanghoon Lee**, Kyung-Joon Park.
IEEE Transactions on Instrumentation and Measurement (TIM), May 2025.
https://ieeexplore.ieee.org/document/10991965

[10] **Cyber-Physical AI. Systematic Research Domain for Integrating AI and Cyber-Physical Systems**
Sanghoon Lee, Jiyeong Chae, Kyung-Joon Park.
ACM Transactions on Cyber-Physical Systems (TCPS), Feb. 2025.
https://dl.acm.org/doi/10.1145/3721437

[9] **Priority-Driven Resource Allocation with Reuse for Platooning in 5G Vehicular Network**
Tae-Woo Kim, **Sanghoon Lee**, Dong-Hyung Lee, Kyung-Joon Park.
Sustainability, Feb. 2025.
https://www.mdpi.com/2071-1050/17/4/1747

[8] **Learning-enabled Flexible Job-Shop Scheduling for Scalable Smart Manufacturing**
Sihoon Moon, **Sanghoon Lee**, Kyung-Joon Park.
Journal of Manufacturing Systems (JMS), Sep. 2024.
https://www.sciencedirect.com/science/article/pii/S0278612524002139

[7] **False Alarm Prevention through Domain Knowledge-driven Machine Learning. Leakage Detection in Water Distribution Networks**
Sanghoon Lee*, Jiyeong Chae*, Sihoon Moon, Kyung-Joon Park.
IEEE Sensors Journal, Aug. 2024.
https://ieeexplore.ieee.org/abstract/document/10643479/

[6] **Learning-Enabled Network-Control Co-design for Energy-Efficient Industrial Internet of Things**
Sihoon Moon, **Sanghoon Lee**, Wonhong Jeon, Kyung-Joon Park.
IEEE Transactions on Network and Service Management (TNSM), Apr. 2024.
https://ieeexplore.ieee.org/abstract/document/10285130

[5] **Deep Reinforcement Learning-driven Scheduling in Multijob Serial Lines. A Case Study in Automotive Parts Assembly**
Sanghoon Lee, Jinyoung Kim, Yongsoon Eun, Kyung-Joon Park.
IEEE Transactions on Industrial Informatics (TII), Feb. 2024.
https://ieeexplore.ieee.org/document/10210628

[4] **A Survey and Perspective on Industrial Cyber-Physical Systems (ICPS). From ICPS to AI-augmented ICPS**
Jiyeong Chae*, **Sanghoon Lee***, Junhyung Jang, Seohyung Hong, Kyung-Joon Park.
IEEE Transactions on Industrial Cyber-Physical Systems (TICPS), Oct. 2023.
https://ieeexplore.ieee.org/abstract/document/10285426

[3] **Graph-based Reinforcement Learning for Flexible Job Shop Scheduling with Transportation Constraints**
Sihoon Moon, **Sanghoon Lee**, Kyung-Joon Park.

Annual Conference of the IEEE Industrial Electronics Society (IECON) 2023, Oct. 2023.
https://ieeexplore.ieee.org/document/10312647

[2] **Real-Time Controller Reconfiguration for Delay-Resilient Cyber-Physical Systems**
Sangjun Kim, **Sanghoon Lee**, Kyung-Joon Park.
IEEE Access, Oct. 2022.
https://ieeexplore.ieee.org/abstract/document/9896851

[1] **Resilient Controller Reconfiguration under Network Delay Attack in Cyber-Physical Systems**
Sangjun Kim, **Sanghoon Lee**, Kyung-Joon Park.
International Conference on Electronics, Information, and Communication (ICEIC) 2022.

DOMESTIC PUBLICATIONS

[D4] **End-to-End Latency Analysis of Reliable Communication in ROS 2 over Wireless Multi-hop Networks**
Taehyun Kim, **Sanghoon Lee**, Kyung-Joon Park.
Korean Institute of Communication and Information Sciences (KICS) Summer Conference, Jun. 2025. *Best Paper Award.*

[D3] **Channel Modeling and SDN-based Routing Path Design for Satellite-Terrestrial Integrated Networks**
Hyunho Ryu, **Sanghoon Lee**, Kyung-Joon Park.
Korean Institute of Communication and Information Sciences (KICS) Summer Conference, Jun. 2024.

[D2] **SDN Based Multipath Routing Algorithm in Satellite-Terrestrial Integrated Network**
Haewon Jeon, **Sanghoon Lee**, Dong-Hyung Lee, Changhee Joo, Kyung-Joon Park.
Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference, Apr. 2024.

[D1] **Sound-based Machine Fault Diagnosis via Active Learning in Industrial Environments with Labeling Constraints**
Jiyeong Chae, **Sanghoon Lee**, Kyung-Joon Park.
Korean Institute of Communication and Information Sciences (KICS) Winter Conference, Jan. 2024.

BOOK IN PREPARATION

ROS 2 System Handbook. Standard reference textbook covering ROS 2 systems engineering, DDS / Zenoh middleware, QoS design, observability, multi-robot integration, and domain playbooks. Target form. hybrid online + offline publication (GitBook / mdBook / GitHub Pages).

PATENTS

[P3] **Method and Apparatus for Generating Indoor Maps**
Jiyeong Chae, Hyunkyo Seo, **Sanghoon Lee**, Kyung-Joon Park.
Republic of Korea, Application No. 10-2024-0145250, Oct. 22, 2024. *Filed.*

[P2] **Method and Apparatus for Detecting Leakage in Water Pipeline Networks**
Young-gi Hong, Kyung-Joon Park, **Sanghoon Lee**, Jiyeong Chae.
Republic of Korea, Application No. 10-2025-0008593, Jan. 21, 2025.

[P1] **Leakage Detection Device and Method**
Kyung-Joon Park, **Sanghoon Lee**, Jiyeong Chae.
Republic of Korea, Application No. 10-2024-0004960, Jan. 11, 2024.

HONORS AND AWARDS

[9] **Early-Career Researcher Academic Research Grant**, Korean Institute of Communication and Information Sciences (KICS). 2026.

[8] **Best Paper Award**, 2025 KICS Summer Conference. 2025.

[7] **Next-Generation Engineering Researcher**, Institute for Promotion of Engineering and Science of Korea (IPESK). 2025.

[6] **Excellence Award in Future Tech Division**, Aero Future Tech & Business Idea Contest, Hanwha Aerospace. *Team Leader*. 2024.

[5] **Selected as 2023 Top 100 Excellent National R&D Achievements**, Ministry of Science and ICT (MSIT), Republic of Korea. *Participant Researcher*. 2023.

[4] **ACM MobiHoc Student Grant**, ACM SIGMOBILE. 2022.

[1–3] **Dean's List (Academic Excellence Award)**, DGIST. Fall 2021, Spring 2021, Spring 2020.

INVITED TALKS AND CONFERENCE PRESENTATIONS

Upcoming (Confirmed)

Date	Venue	Type
2026.10.14 – 16	ICTC 2026, Sheraton Jeju, South Korea	**Invited Talk** — "ROS 2. A Robot Middleware for Physical AI" (tentative)
2026.09.22 – 24	ROSCon 2026, Toronto, Canada	**20-min Talk** — "ROS 2 Middleware Survival Guide. Navigating Space, Time, and State" (#159, accepted)
2026.07.20	Medisby HQ, Seoul, South Korea	Invited Seminar — Physical AI for Rehabilitation Robotics

Past

Date	Venue	Type
2026.03.23	DOGU	Invited Seminar — ROS 2 Architecture Optimization for Multi-Robot Systems
2026.03.09	KETI	Invited Seminar — Multi-Robot + Fleet Management Middleware (DDS, QoS, observability)
2026.01.28	Holiday Robotics	Invited Seminar
2026.01.21	ROSCon KR 2026, Seongnam	Session Speaker

SERVICE TO ROBOITICS COMMUNITY

ROSCon Korea 2027 Organizing Committee Member. Co-host institutions PinkLAB · XYZ · DGIST. Co-chair Prof. Kyung-Joon Park (DGIST). Held at DGIST, Feb. 17–19, 2027. Target attendance 300. Site https://roscon.ros.org/kr/2027/

CONFERENCES ATTENDED

2026 — AW 2026 (Seoul) · ROSCon KR 2026 (Session, Seongnam) · IEEE INFOCOM 2026 (Workshop, Tokyo, Japan) · KICS Summer 2026 (Jeju) · ICROS 2026 (Daegu) · IFAC 2026 (Busan, scheduled) · ROSCon 2026 (Toronto, scheduled) · IROS 2026 (Pittsburgh, scheduled) · ICTC 2026 (Jeju, invited talk scheduled) · Middleware 2026 (Tarragona, conditional)

2025 — AIoT 2025 (Seoul) · ICCAS 2025 (Poster, Incheon) · ROSCon 2025 (Singapore) · CoRL 2025 (Seoul) · IEEE ICNP 2025 (Poster, Seoul) · ICML 2025 (Vancouver) · ICUFN 2025 (Workshop Paper, Lisbon) · CISC-S

2025 (Kyeungju) · KICS Summer 2025 (Poster, Jeju) · Joint Workshop on Physical AI (South Korea) · AW 2025 (Seoul)

2024 — AIoT 2024 (Seoul) · AW 2024 (Seoul) · JCCI 2024 (Busan) · ROSCon 2024 (Odense, Denmark) · KICS Winter 2024 (PyeongChang) · KICS Summer 2024 (Jeju)

2023 — ICML 2023 (Hawaii, USA) · KICS Summer 2023 (Jeju)

2022 — ICEIC 2022 (Jeju) · ACM MobiHoc 2022 (Seoul) · ICROS 2022 (Geoje)

PROFESSIONAL SERVICE — PEER REVIEW

2026 – Present
- *Springer Nature*. Artificial Intelligence Review. Computing
- *IEEE*. Transactions on Systems, Man, and Cybernetics

2025 – Present
- *Springer Nature*. Scientific Reports. International Journal of Computational Intelligence Systems
- *IEEE*. Journal of Communications and Networks (JCN). Systems Journal. Transactions on Network and Service Management (TNSM)
- *Elsevier*. Expert Systems With Applications. Computers and Operations Research

2024 – Present
- *Springer*. Machine Learning
- *Elsevier*. ICT Express
- *IEEE*. Transactions on Industrial Cyber-Physical Systems (TICPS)

2023 – Present
- *IEEE*. Access

OPEN-SOURCE SOFTWARE

QoS Guard — ROS 2 DDS QoS Static Validator
Offline verification of 16 QoS policies across Discovery / Data Exchange / Disassociation phases with detection of 40 common constraint violations. Companion to publication [U8].
https://github.com/csi-dgist/QoS-Guard · Project Page · arXiv:2509.03381

DDS Optimizer for Wireless Large Payload Transfer — ROS 2 Wireless Communication Optimizer
XML-based QoS configuration framework for ROS 2 DDS over wireless. Caps RTPS message size at 1472 B, optimizes heartbeat rates, and sizes HistoryCache to prevent buffer bursts. Companion to publication [U6].
https://github.com/csi-dgist/DDS-Optimizer-for-Wirelss-Large-Payload-Transfer · arXiv:2508.11366

ros2probe — Host-Level Observability for ROS 2 DDS Traffic
eBPF-based passive monitoring tool that captures RTPS/DDS packets and reconstructs the full ROS graph, topic metrics, and message streams in userspace, with zero subscriptions and no middleware intrusion. SEDP event-driven data-centric GID synchronization (C3) and adaptive visibility (C4) provide non-trivial novelty. Rust implementation. CLI `rp` / GUI / MCAP recorder / Unix-socket JSON-RPC. Validated on Laptop, NVIDIA Jetson Orin series, Raspberry Pi 4/5. Companion to publication [U11].
https://github.com/csi-dgist/ros2probe · Project Page

GitHub cumulative stars 100+.

TECHNICAL SKILLS

Systems — C++, Python, Rust, Linux (Kernel / Driver Analysis), eBPF, Docker, Git
Middleware — ROS 2, DDS (FastDDS, CycloneDDS, RTI Connext), Zenoh, TCP/IP, UDP
Intelligence — PyTorch, TensorFlow, Nav2, MoveIt 2, Cartographer, AMCL
Simulation — NVIDIA Isaac Sim, MuJoCo, Gazebo, MATLAB, NS-3, OMNeT++

CITATIONS AND INDICES

Google Scholar. 200+ citations (Apr. 2026). h-index 8. i10-index 5. Citation milestones. 100 reached Sep. 8, 2025. 200 reached Apr. 23, 2026.

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